

Step 1: Install Django & Django REST Framework

Open terminal and run:

```
# Create virtual environment
python -m venv myenv

# Activate virtual environment

# Windows
myenv\Scripts\activate

# Mac/Linux
source myenv/bin/activate

# Install Django and DRF
pip install django djangorestframework
```

Now create project and app:

```
# Create Django project
django-admin startproject foodiehub

# Go inside project folder
cd foodiehub

# Create app
python manage.py startapp api
```

Step 2: Add REST Framework to INSTALLED_APPS

Open:

```
foodiehub/settings.py
```

Find:

```
INSTALLED_APPS = [
```

Add:

```
'rest_framework',  
'api',
```

Example:

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
  
    'rest_framework',  
    'api',  
]
```

Now run server:

```
python manage.py runserver
```

If everything is correct, server will start without errors.

Step 3: Create Function-Based API Endpoint

Open:

```
api/views.py
```

Write:

```
from rest_framework.decorators import api_view  
from rest_framework.response import Response  
  
@api_view(['GET'])  
def hello_spotify(request):  
    return Response({
```

```
        "message": "Hello, Spotify Fans!"
    })
```

```
# -----
# JSON vs XML Example (Flipkart Product)
#
# JSON Example:
# {
#     "name": "iPhone 15",
#     "price": 79999
# }
#
# XML Example:
# <product>
#     <name>iPhone 15</name>
#     <price>79999</price>
# </product>
#
# JSON is lightweight and easy to read.
# XML uses tags and is more verbose.
# -----
```

Step 4: Create URL for API

Create file:

```
api/urls.py
```

Add:

```
from django.urls import path
from .views import hello_spotify

urlpatterns = [
    path('hello_spotify/', hello_spotify),
]
```

Now open:

```
foodiehub/urls.py
```

Modify:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('api/', include('api.urls')),
]
```

Now visit:

```
http://127.0.0.1:8000/api/hello_spotify/
```

Output:

```
{
  "message": "Hello, Spotify Fans!"
}
```

Step 5: Create Serializer

Create file:

```
api/serializers.py
```

Write:

```
from rest_framework import serializers

class RestaurantSerializer(serializers.Serializer):
    name = serializers.CharField(max_length=100)
    cuisine = serializers.CharField(max_length=100)
```

Example restaurant data:

```
restaurant = {  
  "name": "Spice Garden",  
  "cuisine": "Indian"  
}
```

This serializer converts Restaurant objects into JSON format and validates data.